

Change Management in PLM: A Practical Guide

Sample document written for RAG practice. General information only.

Why change management matters

Product lifecycle management (PLM) systems hold the definitive record of a product: parts, documents, bills of materials and the relationships between them. When an engineer wants to modify any of these, the change has to be controlled. Without control, manufacturing may build from an outdated drawing, suppliers may quote against the wrong revision, and service teams may stock parts that no longer fit. A formal change process keeps every department working from the same, approved version of the truth.

The change lifecycle

Most organisations follow three stages. First, a problem report describes an issue, such as a part that fails a test or a drawing error found on the shop floor. Second, a change request proposes a solution and lists the affected items. Third, a change notice implements the approved solution: it defines exactly which objects are revised and when the new revisions take effect. Each stage has an owner, a due date and a defined set of reviewers.

Roles and approvals

A typical change board includes design engineering, manufacturing engineering, quality, purchasing and, for regulated products, compliance. Reviewers can approve, reject or ask for more information. A change should not be released until every mandatory reviewer has signed off. Delegating approvals during holidays is a common source of delays, so many companies configure backup approvers in advance.

Revisions and versions

A version records a saved state of an object while it is being worked on. A revision marks a released, approved state. Working copies can have many versions, but only released revisions should reach manufacturing. When a revision is superseded, the old revision remains in the database for traceability. Teams should never delete released data; they should obsolete it.

Common mistakes

Frequent problems include vague change descriptions, missing impact analysis, and changes released without updating the bill of materials. Another mistake is treating the process as paperwork. The best teams use the impact analysis to ask what else this change touches: assemblies, tooling, software, packaging, and spare parts. A good change record answers the questions people will ask two years later.